

Multiresolution Gray-Scale and Rotation Invariant Texture Classification with Local Binary Patterns

http://www.scholarpedia.org/article/Local_Binary_Patterns

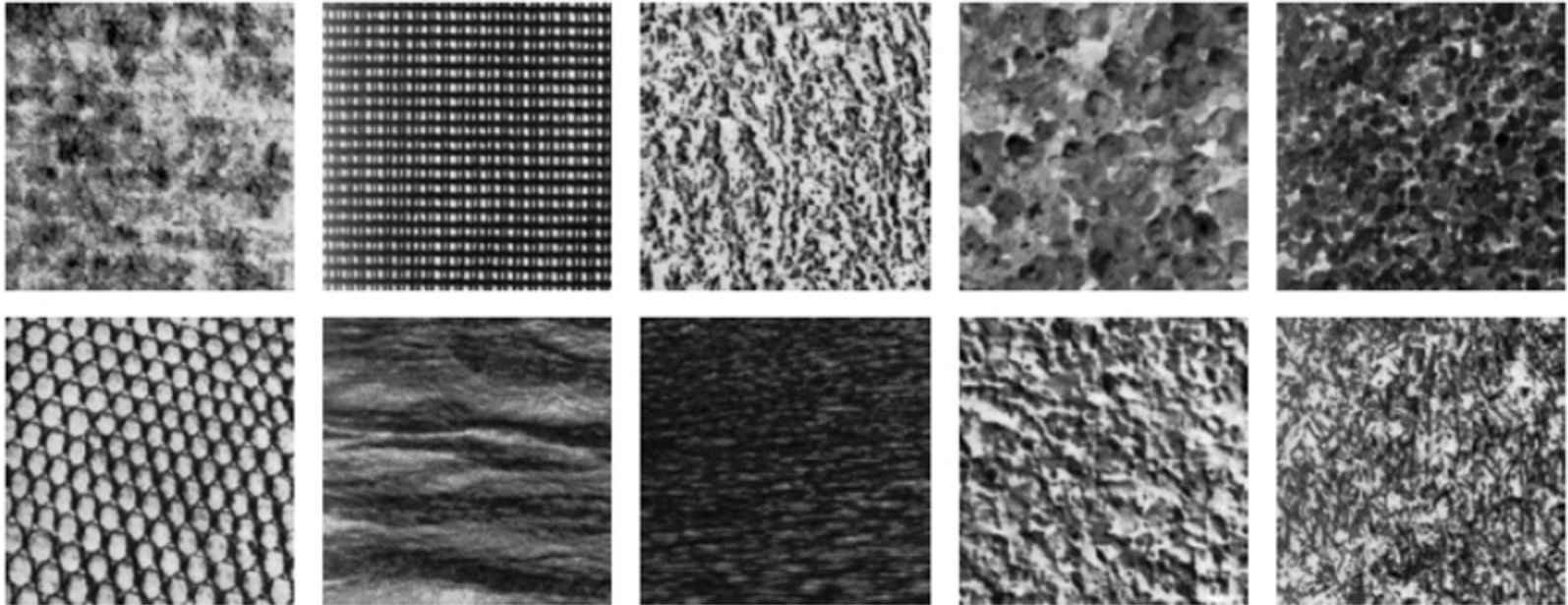
https://en.wikipedia.org/wiki/Local_binary_patterns

T. Ojala, M. Pietikäinen & T. Mäenpää,
IEEE Trans. PAMI (2002)

Arthur K. Tang and Chris Z. Yu

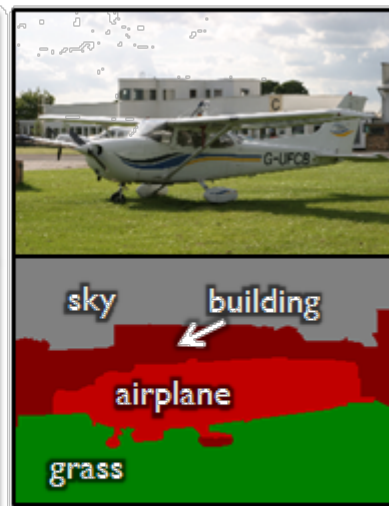
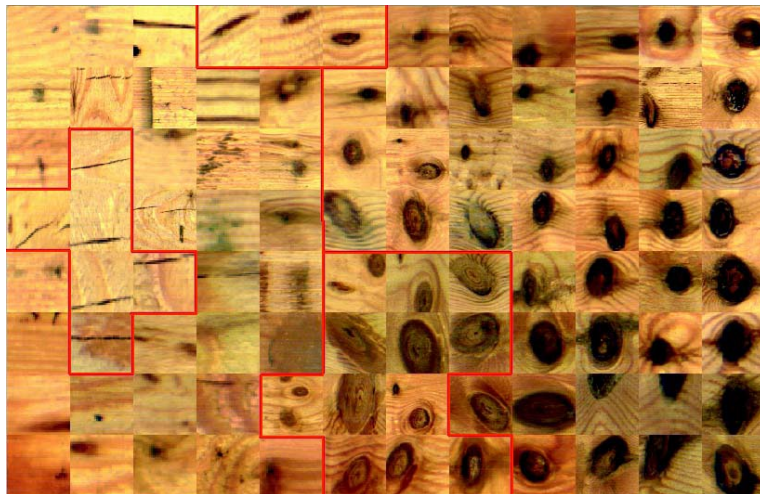
Modified by Albert Chung and Jierong Wang

Introduction



Classification refers to assigning a physical object or an incident to one of a set of predefined categories. In texture classification, the goal is to assign an unknown sample image to one of a set of known texture classes.

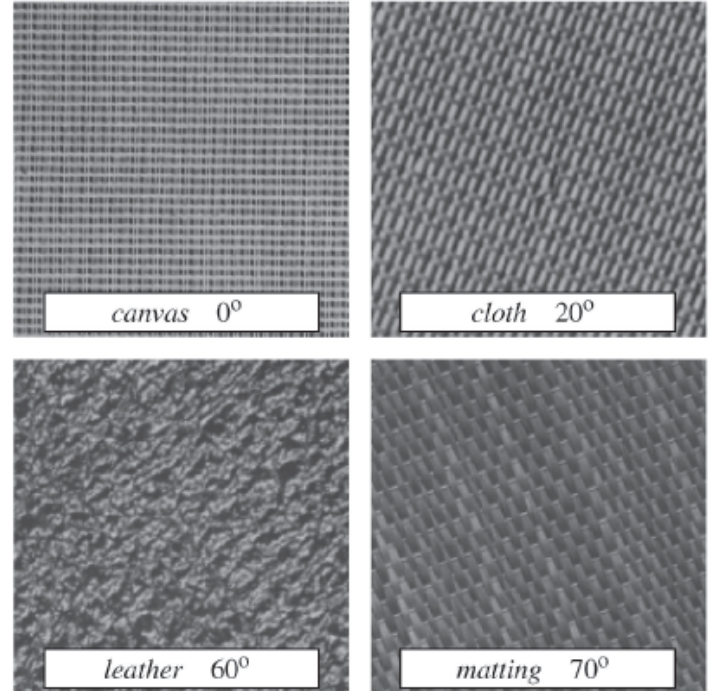
Introduction



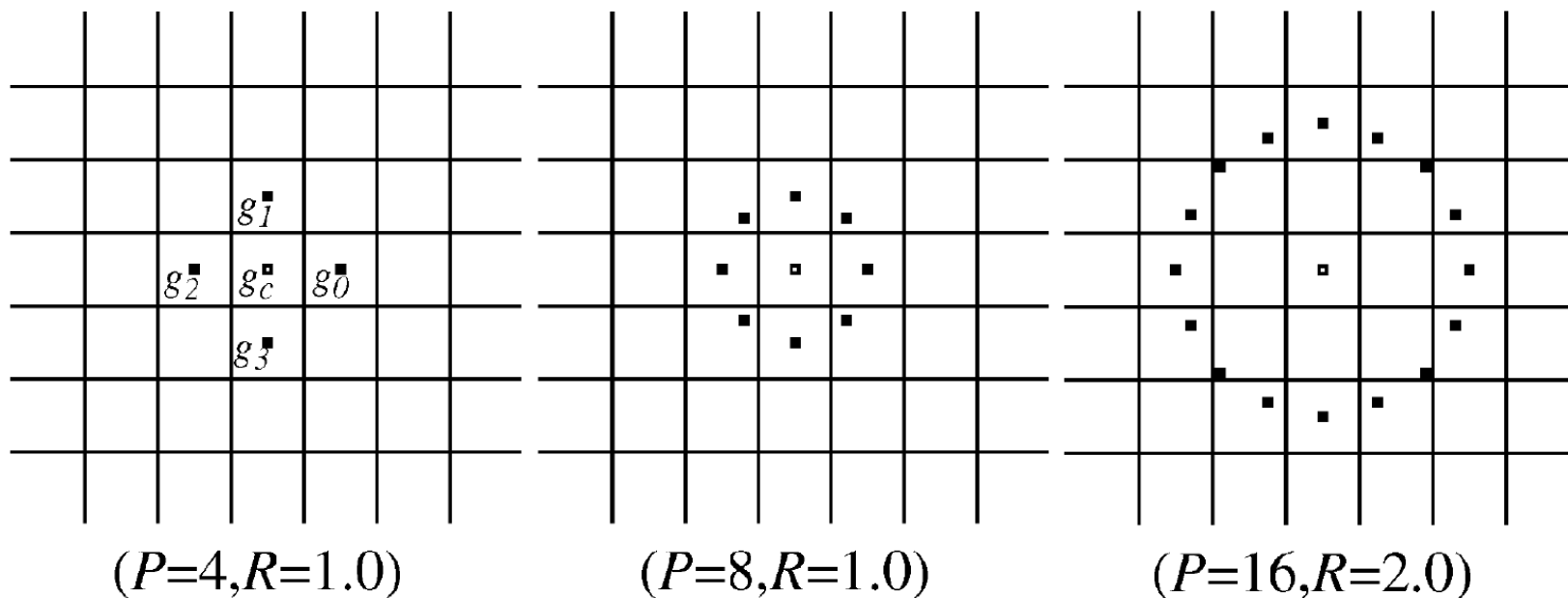
object classes	building	grass	tree	cow	sheep	sky	airplane	water	face	car
bicycle	flower	sign	bird	book	chair	road	cat	dog	body	boat

Introduction

- Local Binary Patterns (LBP) is
 - Invariant to grayscale variation
 - Invariant to rotation
 - Robust and computationally efficient for real applications



Feature Descriptor: Local Binary Pattern (LBP)



P : Size of the neighboring pixel set

R : Radius of the neighboring pixel circle

g_c : center pixel grayscale value

g_p : gray level of the p^{th} neighboring pixel ($p = 0, 1, \dots, P-1$)

Feature Descriptor: Local Binary Pattern (LBP)

Fig. 4.3 Neighborhood set for different (P, R) . The pixel values are bilinearly interpolated whenever the sampling point is not in the center of a pixel

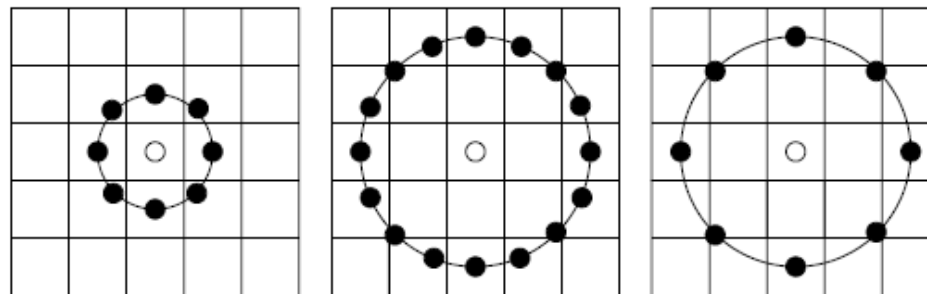
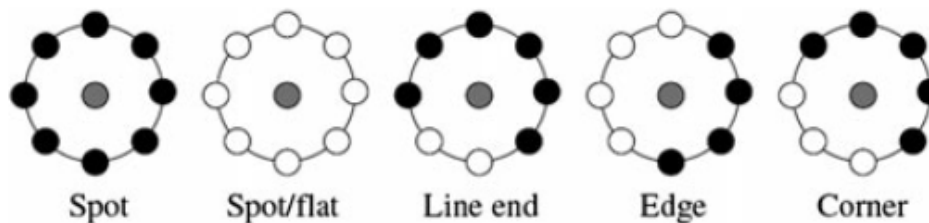


Fig. 4.4 Examples of texture primitives detected by LBP (white circles represent ones and black zeros)



P : Size of the neighboring pixel set

R : Radius of the neighboring pixel circle

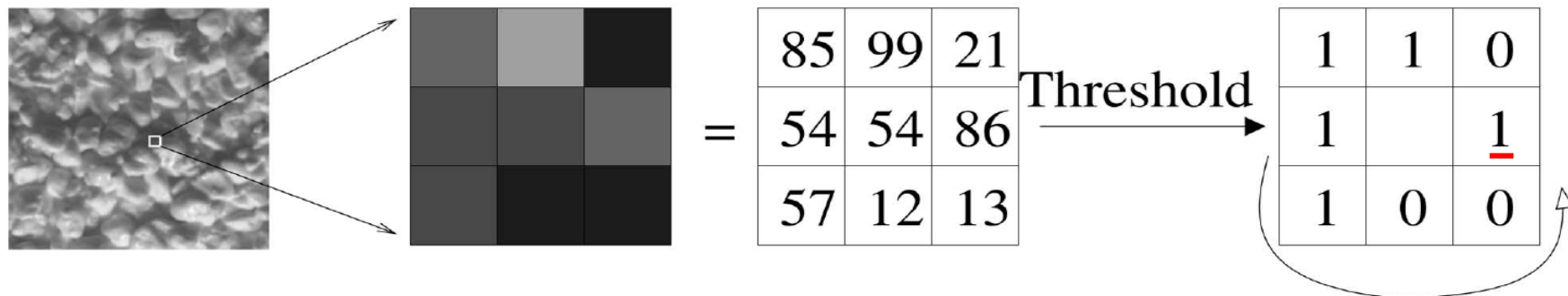
g_c : center pixel grayscale value

g_p : gray level of the p^{th} neighboring pixel ($p = 0, 1, \dots, P-1$)

Feature Descriptor: Local Binary Pattern (LBP)

$$LBP_{P,R} = \sum_{p=0}^{P-1} s(g_p - g_c) 2^p.$$

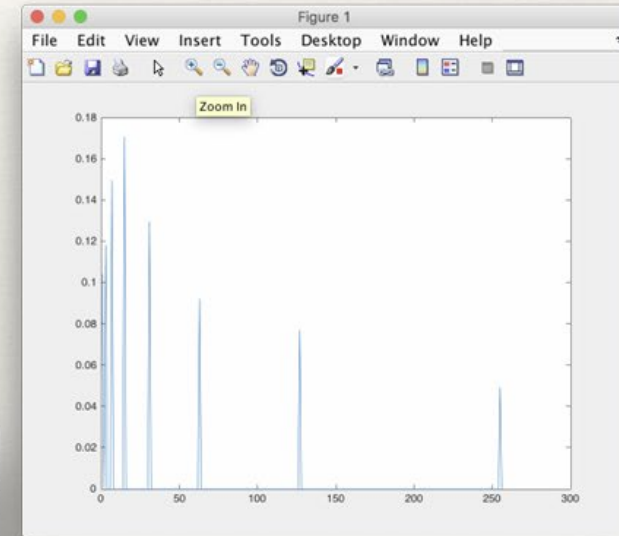
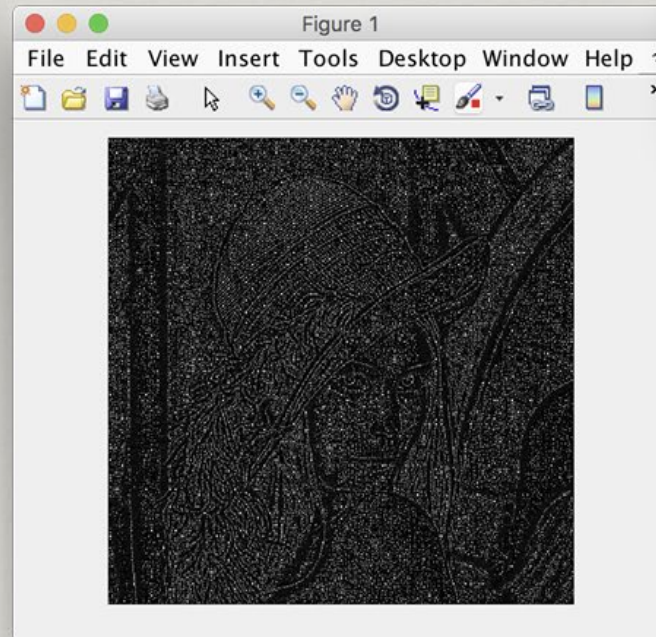
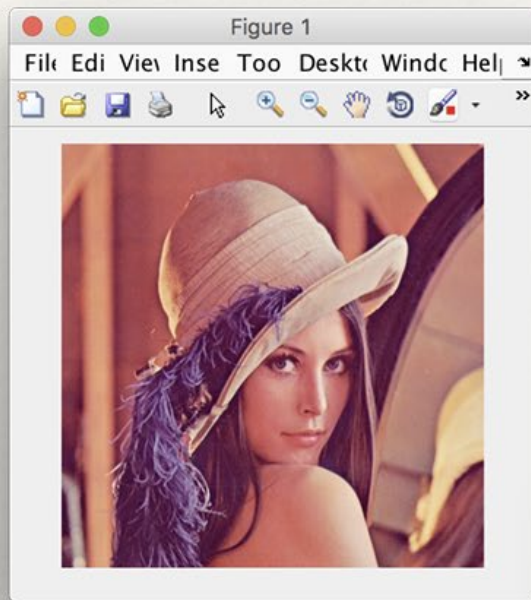
$$s(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0. \end{cases}$$



Binary representation: 00111101 (8 bits)

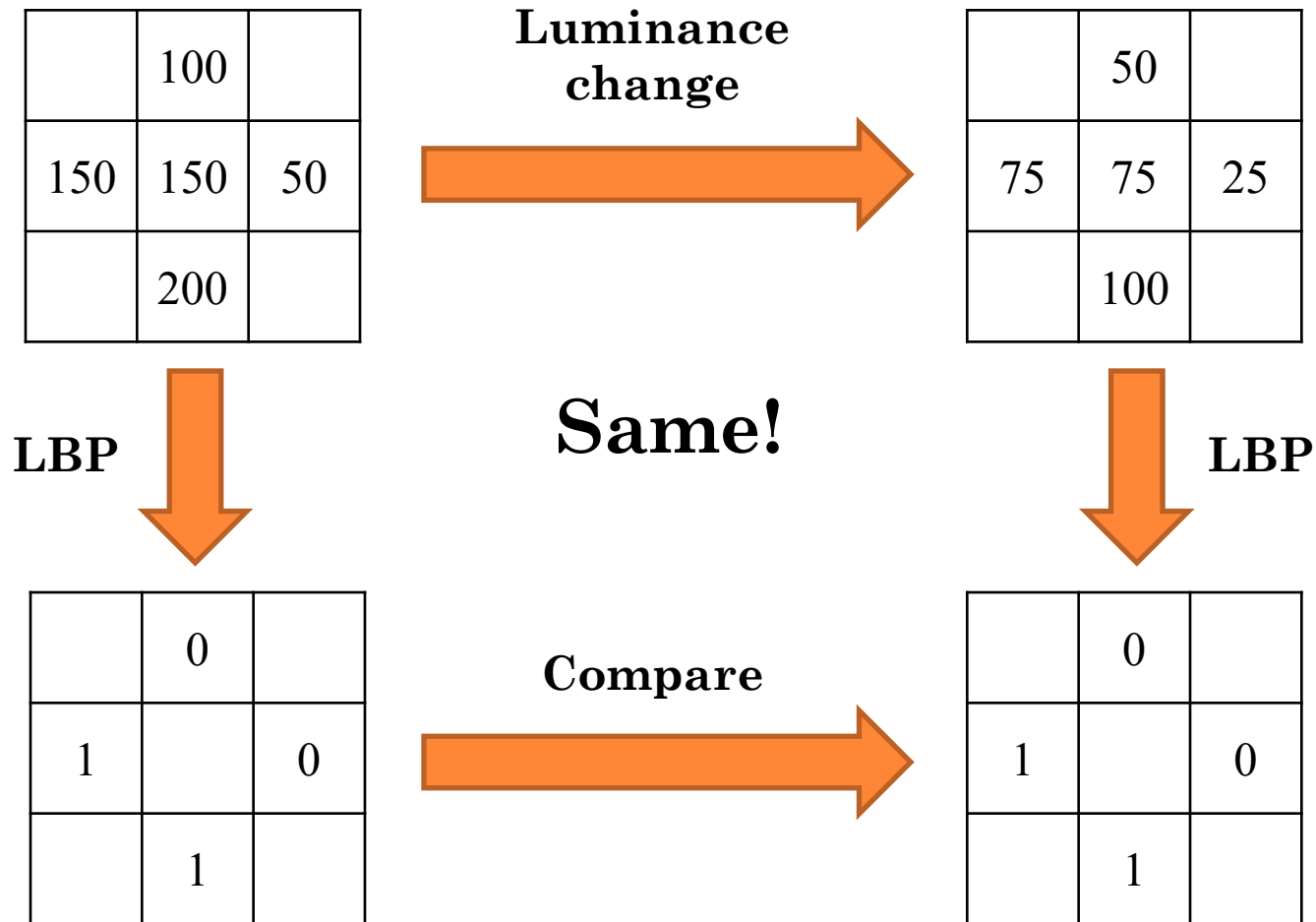
Decimal representation: 61

Feature Descriptor: Local Binary Pattern (LBP)



- ❖ For whole image
- ❖ For each 16×16 cell in the image

Why is LBP Grayscale Shift Invariant?

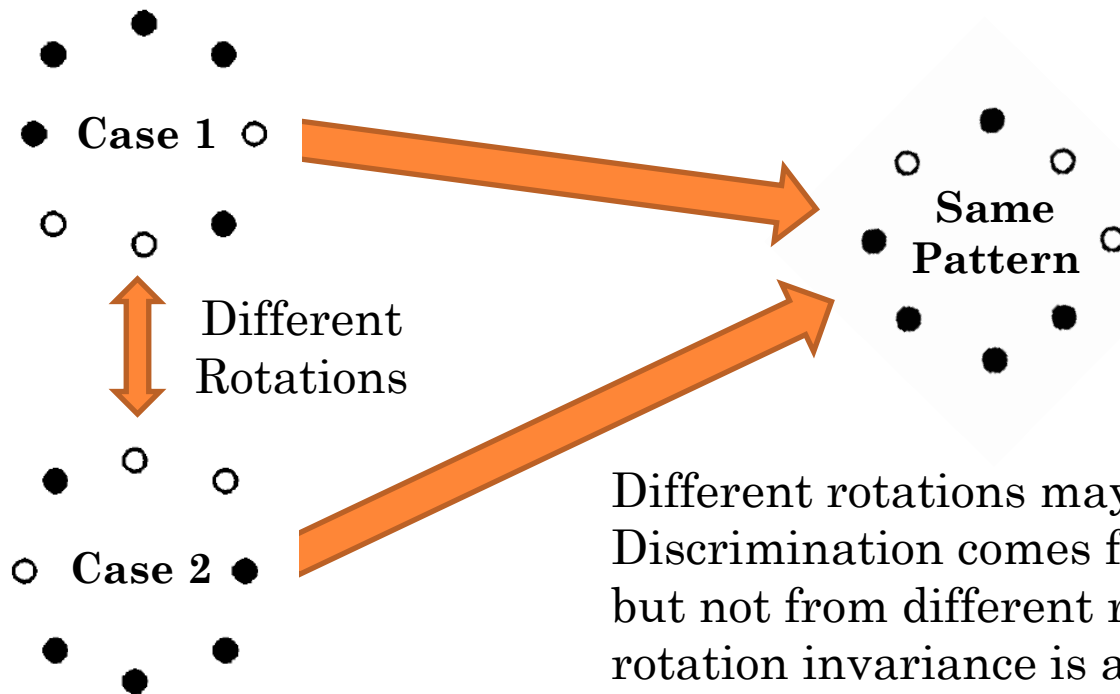


Why is LBP Rotation Invariant?

Performs circular bit-wise right shift on the P -bit number i times so that the LBP value is minimized.

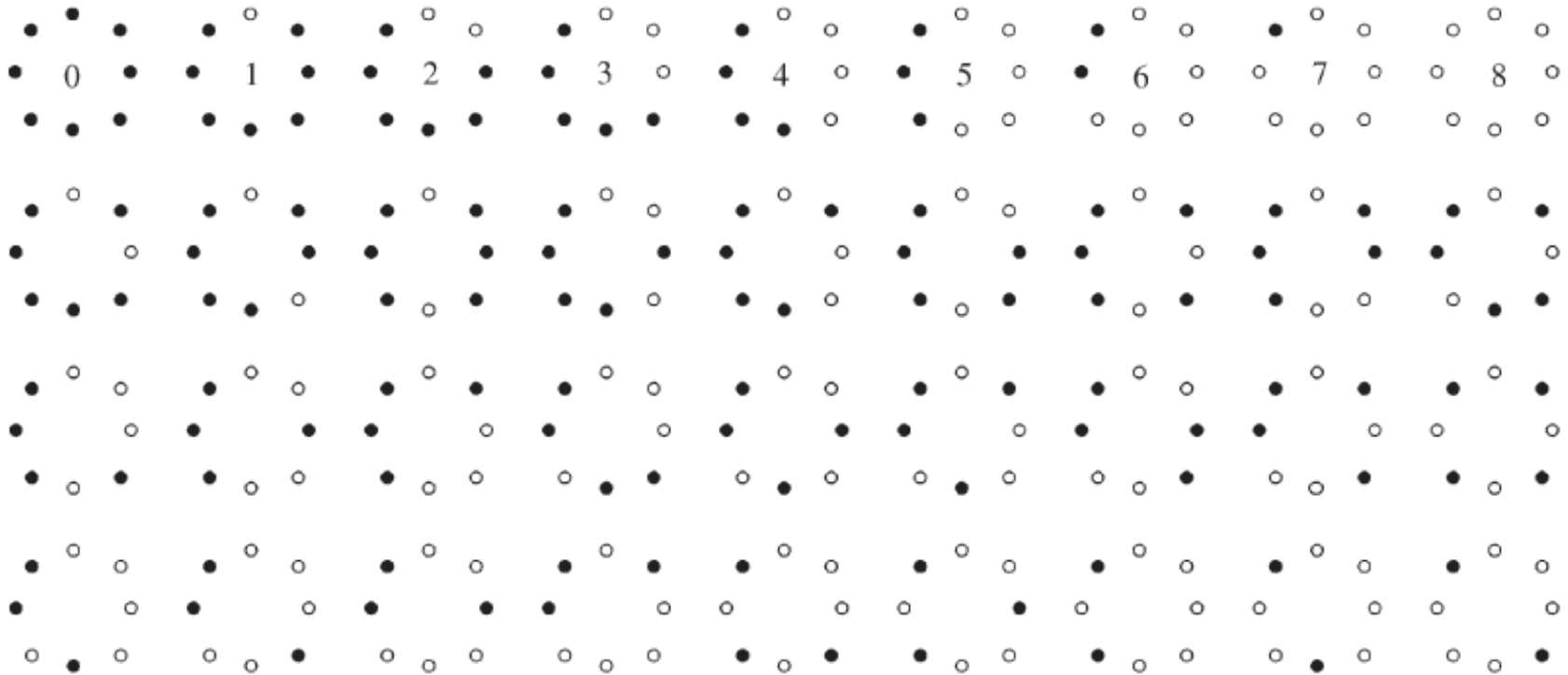
The purpose is to unify different cases with the same spatial pattern.

$$LBP_{P,R}^{ri} = \min\{ROR(LBP_{P,R}, i) \mid i = 0, 1, \dots, P - 1\},$$



Different rotations may give the same pattern. Discrimination comes from different patterns, but not from different rotations. This is how rotation invariance is achieved.

Rotation Invariant Patterns



Number of Cases: $2^8 = 256$ possible patterns

Number of Patterns: 36 (rotation invariant patterns)

Uniform Patterns

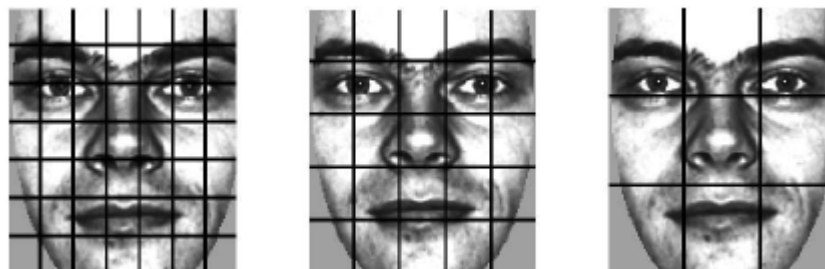


$$LBP_{P,R}^{riu2} = \begin{cases} \sum_{p=0}^{P-1} s(g_p - g_c) & \text{if } U(LBP_{P,R}) \leq 2 \\ P + 1 & \text{otherwise,} \end{cases}$$

$$U(LBP_{P,R}) = | s(g_{P-1} - g_c) - s(g_0 - g_c) | + \sum_{p=1}^{P-1} | s(g_p - g_c) - s(g_{p-1} - g_c) | .$$

- The chance of the occurrence of the 9 uniform patterns can go up to more 90%.
- Only Uniform Patterns are used.
- Other non-uniform patterns are grouped as P+1.
- For the uniform patterns, the values range from 0 to P, based on how many bit '1's exist in the neighboring set.

Application: Face Recognition



Algorithm:

- ① Divide a face image into several regions.
- ② Compute $LBP_{8,2}^{riu2}$ on each region.
- ③ Concatenate all LBP vectors into a long vector ----- face feature.
- ④ Apply PCA or other related dimensionality reduction methods on the face feature.

Reference: T. Ahonen, A. Hadid, and M. Pietikainen, "*Face Description with Local Binary Patterns: Application to Face Recognition*", IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. 28, No. 12, pages 2037-2041, 2006.

More details:

http://www.scholarpedia.org/article/Local_Binary_Patterns

Application: Face Recognition

Different facial feature computation methods.

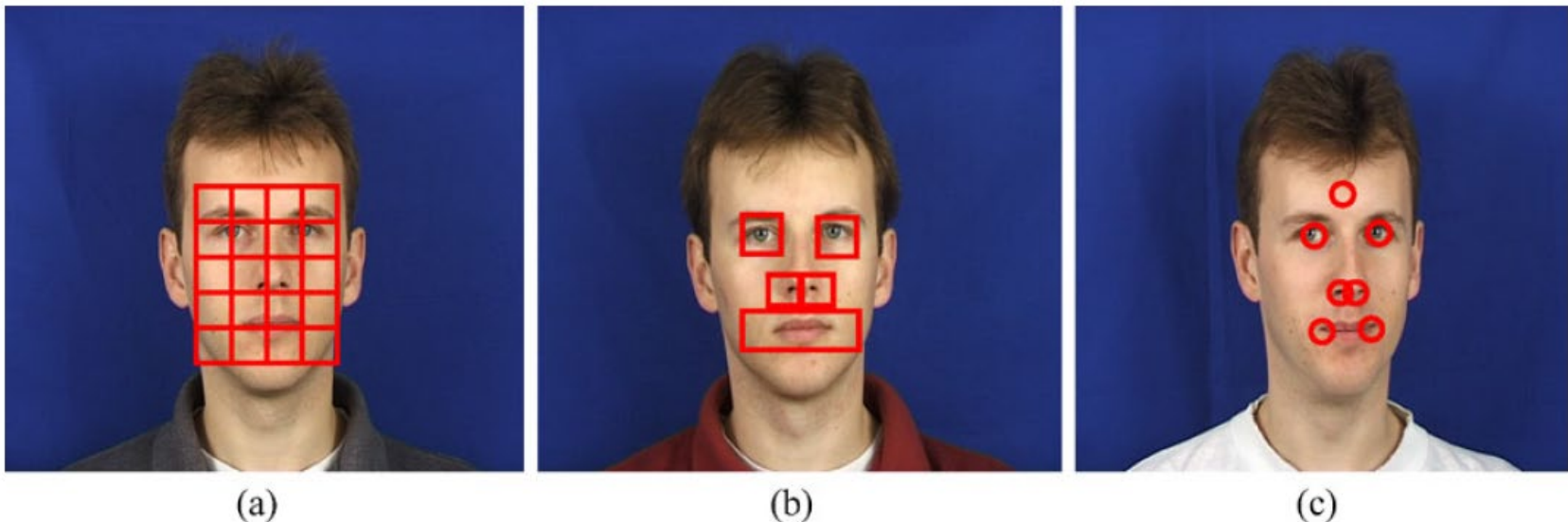
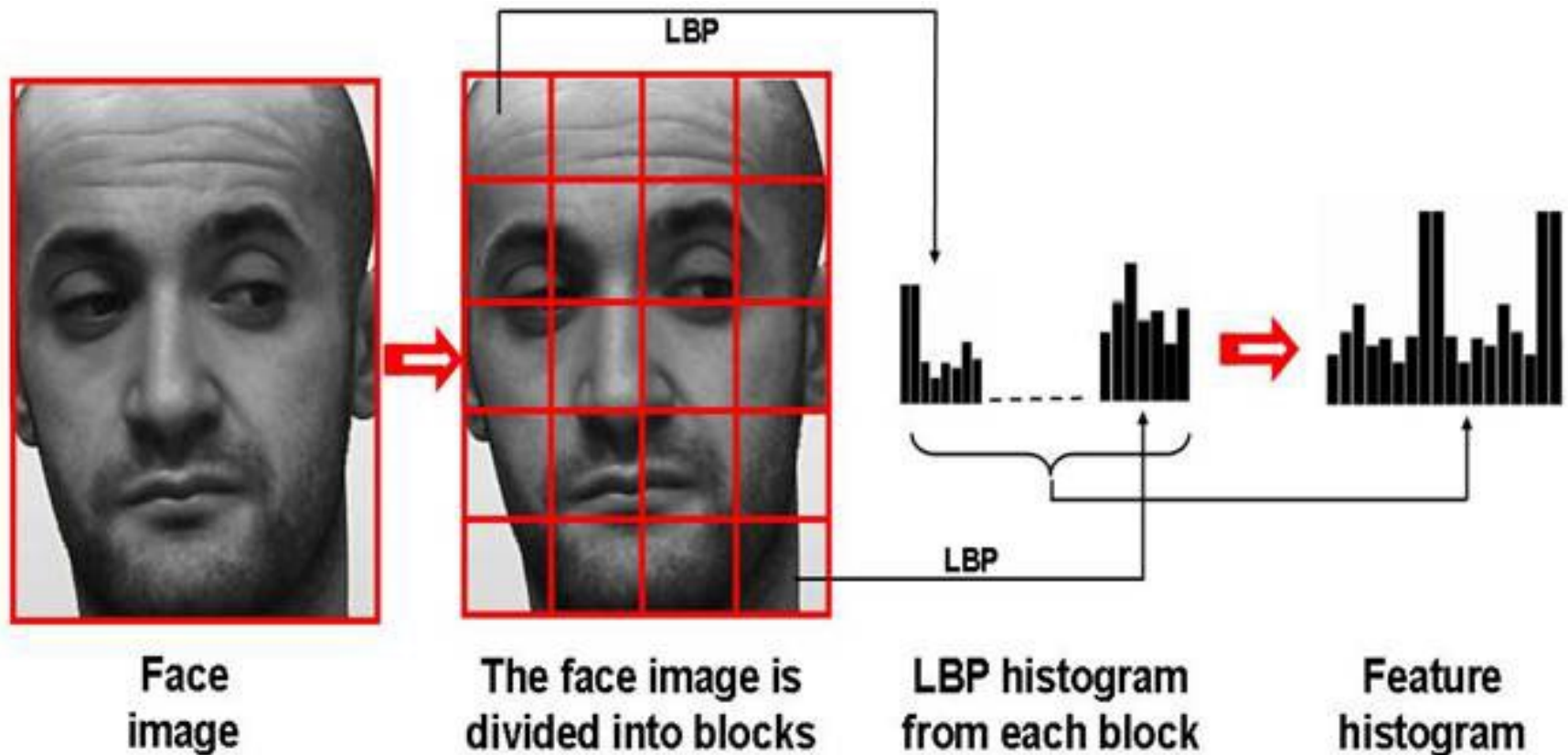


Fig. 4.1 Facial feature computation from **a** a regular grid of fixed size regions, **b** irregular variable size regions (feature-driven) and **c** around central feature locations

Application: Face Recognition



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Application: Face Recognition

LBP based facial representation

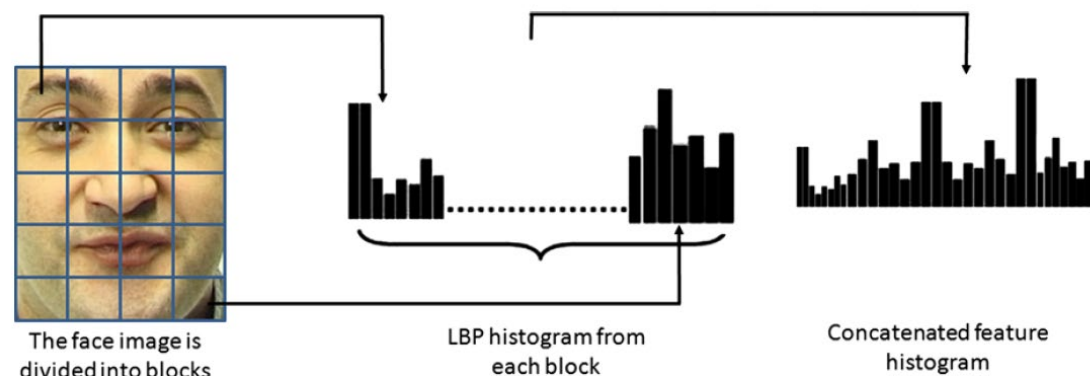


Fig. 4.6 Example of an LBP based facial representation

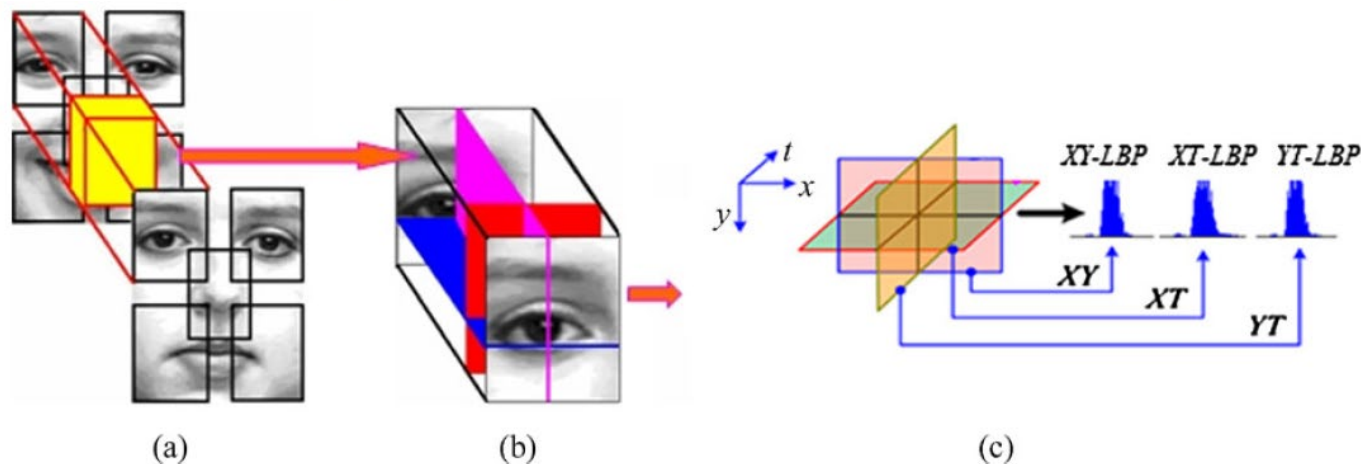


Fig. 4.7 Features in each block volume. **a** Block volumes; **b** LBP features from three orthogonal planes; **c** Concatenated features for one block volume with the appearance and motion

Application: Face Recognition

LBP based facial expression recognition



Fig. 4.10 Selected 15 slices for different facial expression pairs